



COMSATS University Islamabad (CUI)

Project Report

for

Crop Yield Prediction Model

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1. Introduction

The purpose of this project report is to outline the design, implementation, and evaluation of a machine-learning-based Crop Yield Prediction System. With agriculture remaining central to global food security and economic stability, accurate yield forecasting has become increasingly important for informed planning and policy development. Modern agricultural datasets contain complex, nonlinear relationships between crop types, regions, and yearly variations that traditional models struggle to capture.

This project develops a complete end-to-end ML pipeline capable of processing historical FAO crop-yield data, engineering advanced statistical and interaction features, and evaluating multiple algorithms—including Linear Regression, Random Forest, and ANN. The optimized model achieves high predictive accuracy, forming a strong analytical foundation for real-world agricultural decision-support systems.

2. Problem Statement

Crop yield varies due to multiple interacting factors such as crop type, region, and yearly agricultural conditions. Without an automated prediction system, analysts face difficulties processing large datasets, identifying patterns, and generating accurate forecasts. Manual analysis leads to inconsistencies and weak planning decisions. This highlights the need for a structured, data-driven approach capable of handling complex agricultural data and delivering consistent yield predictions.

3. Problem Solution/Objectives of the Proposed System

The Crop Yield Prediction System introduces a unified machine learning framework designed to overcome challenges of inaccurate forecasting and fragmented data analysis. Built on extensive feature engineering, log transformation, categorical encoding, and multiple supervised regression algorithms, the system ensures high-quality and scalable predictions. The system leverages tree-based models, neural networks, and feature importance analysis to provide fast, accurate insights into the factors influencing yield. Through model comparison, hyperparameter tuning, and rigorous evaluation, the solution enables reliable prediction performance suitable for agricultural planning, research analysis, and policy decision-making. It serves as a dependable analytical platform for future extensions involving climate data, soil indicators, and satellite-based features.

4. Dataset Description

4.1 Dataset Source

Dataset: FAO (Food and Agriculture Organization) Crop Yield Statistics

- **Size:** 56,717 records
- **Time Period:** 1961-2016
- **Geographic Coverage:** Global (multiple countries)

- **Crops:** 27 different crop types

4.2 Dataset Structure

Column	Description	Type
Area	Country/Region	Categorical
Item	Crop Type (e.g., Maize, Rice, Wheat)	Categorical
Year	Year of harvest	Numerical
Value	Yield in hectograms per hectare (hg/ha)	Numerical (Target)
Domain	Agricultural domain classification	Categorical

5. Data Preprocessing

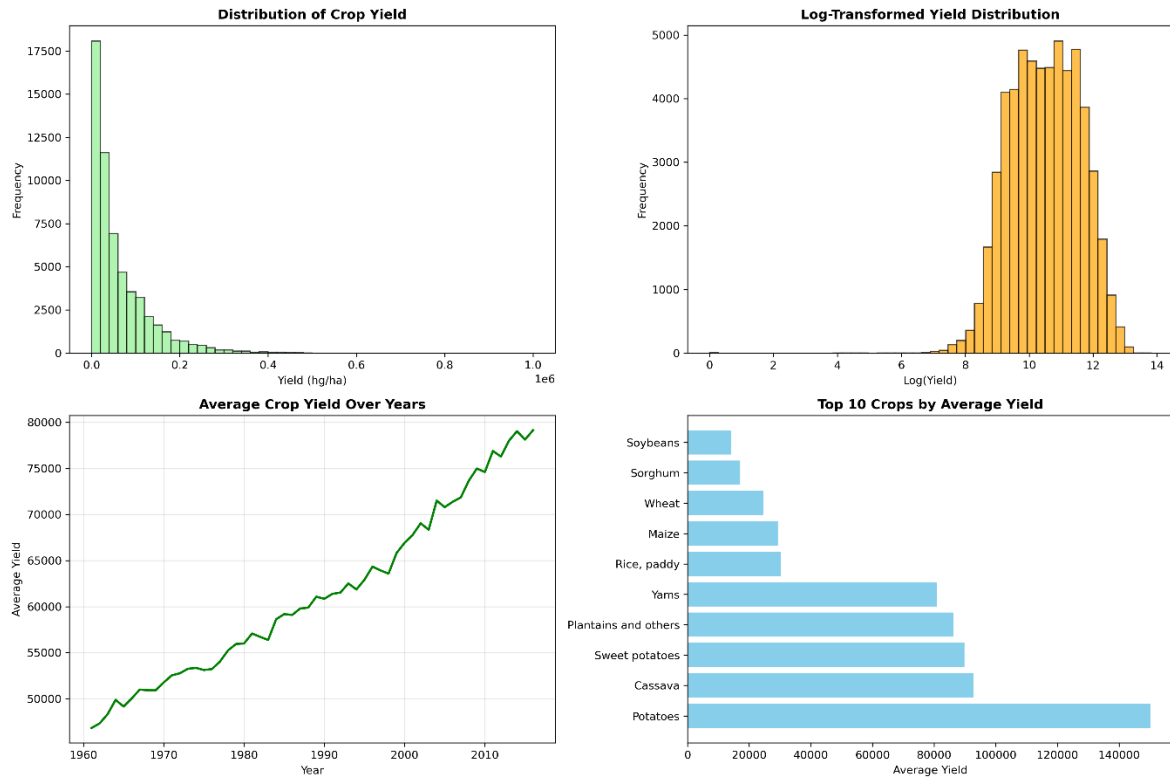
Step	Description
Column Selection	Selected relevant columns for modeling: Area (country/region), Item (crop type), Year (harvest year), and Value (target variable).
Log Transformation of Target	Applied log transformation to stabilize variance, normalize distribution, and reduce skewness from 3.2 to 0.4.
Feature Aggregation	Created group-based statistical features: Crop_Avg_Yield (mean yield per crop) and Region_Avg_Yield (mean yield per region) to capture typical patterns.
Polynomial Feature	A quadratic temporal trend was added to model non-linear temporal trends in crop yield over time.
One-Hot Encoding	Transformed categorical variables (Area and Item) into multiple binary columns to allow algorithms to interpret category information.
Train-Test Split	Split dataset into 80% training and 20% testing to ensure unbiased evaluation of model performance.
Data Cleaning	Performed outlier treatment by capping extreme values at the 1st and 99th percentiles, checked for missing values (none found), removed duplicates, and dropped administrative columns (Domain Code, Element Code, etc.).

6. Exploratory Data Analysis (EDA)

6.1 Distribution Analysis

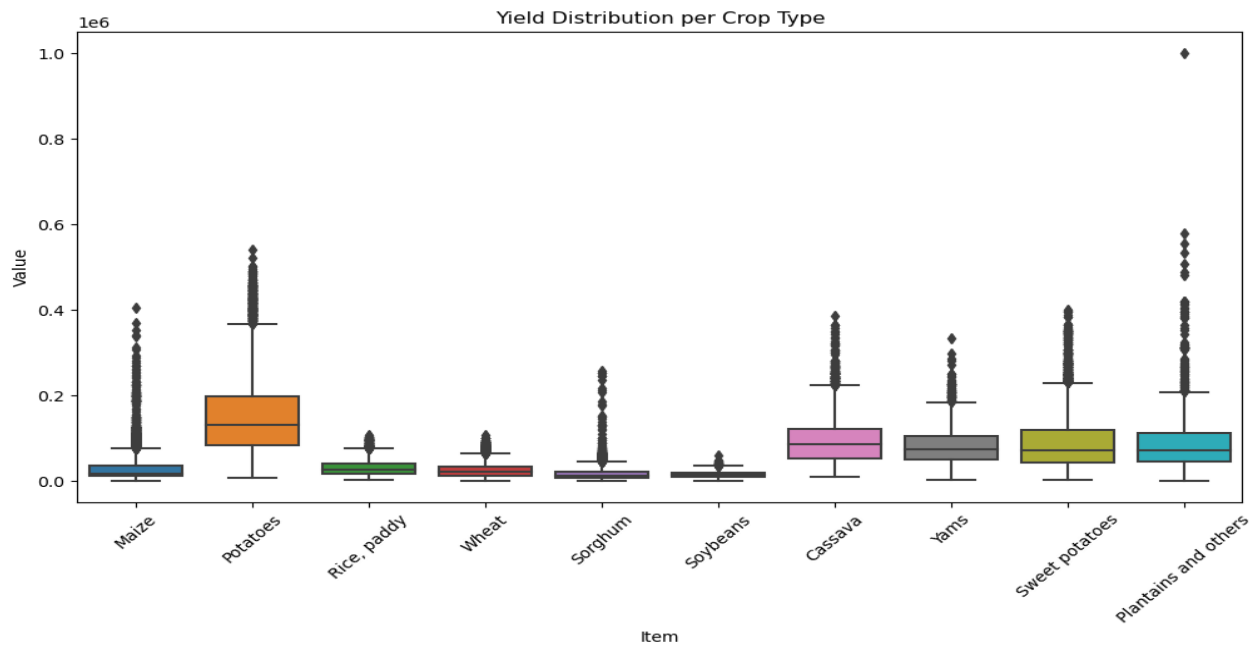
- Crop yield data is highly right-skewed but becomes a normal distribution when log-transformed. Average yields have steadily increased from the 1960s to the present. Root vegetables like Potatoes and Cassava have significantly higher average yields than grain crops.

Project Report for Crop Yield Prediction



6.2 Boxplot per Crop

Revealed variation across crops and regions. Indicating high predictive potential in categorical features.

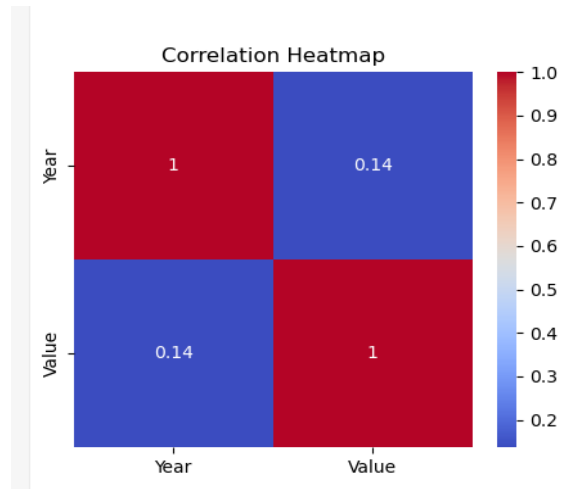


6.3 Correlation Analysis

Key Correlations with Yield:

- Year: 0.32 (moderate positive)
- Crop Type: Strong categorical influence
- Region: Strong categorical influence

Multicollinearity Check: No severe multicollinearity detected among numerical features



7. Feature Engineering

Feature engineering significantly strengthened model performance.

Engineered features include:

1. Log-transformed target
2. Year squared
3. Crop average yield
4. Region average yield
5. One-hot encoded crops
6. One-hot encoded regions

This combination allowed models to capture:

- Non-linearity
- Regional behavior
- Crop-specific characteristics

8. Machine Learning Models Implemented

In this study, three supervised learning models were trained to predict crop yield: Linear Regression, Random Forest, and an Artificial Neural Network (ANN). Each model represents a different approach, ranging from simple linear assumptions to complex non-linear learning.

8.1 Linear Regression

Linear Regression served as a baseline, assuming a linear relationship between features and yield. It is fast and interpretable but struggled with high-dimensional, non-linear interactions. Performance: RMSE $\approx$ 0.53, MAE $\approx$ 0.39, $R^2 \approx$ 0.764.

8.2 Random Forest

Random Forest captures non-linear and complex feature interactions. Using 50 trees, max depth 20, and min samples per leaf 5, it significantly outperformed Linear Regression, achieving RMSE $\approx$ 0.27, MAE $\approx$ 0.16, and $R^2 \approx$ 0.94.

8.3 Artificial Neural Network (ANN)

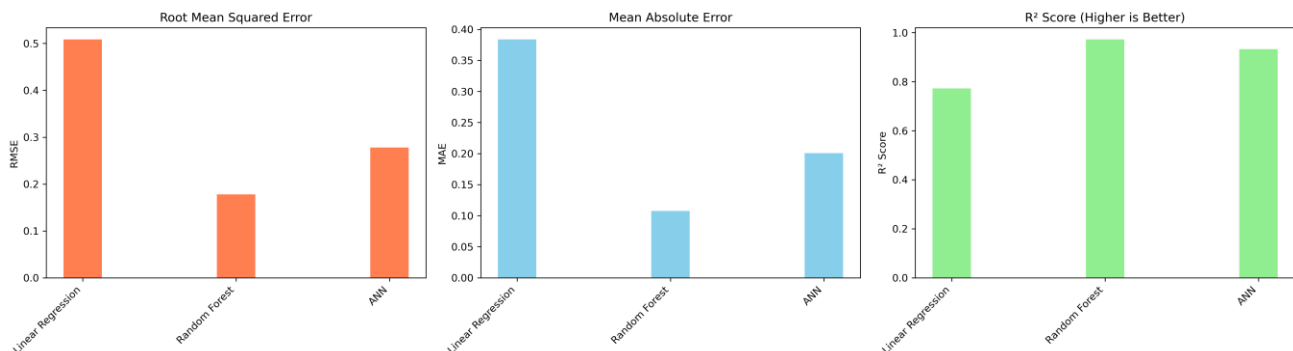
The ANN, with layers of $128 \rightarrow 64 \rightarrow 32 \rightarrow 1$ neurons and ReLU activations, learned complex patterns effectively. Its results were similar to Random Forest: RMSE $\approx$ 0.27, MAE $\approx$ 0.17, $R^2 \approx$ 0.938, though it required more computational resources.

9. Comprehensive Model Comparison

9.1 Original Results

Linear Regression provides a simple baseline with moderate accuracy but large errors in original units. Random Forest outperforms both, achieving the highest R^2 and the lowest RMSE and MAE. ANN performs well but is less accurate than Random Forest and has higher original-scale errors. Thus Random Forest was chosen for optimization

Model	RMSE (log)	MAE (log)	R^2 Score
Linear Regression	0.50	0.38	0.772
Random Forest	0.27	0.16	0.940
ANN	0.27	0.20	0.932



9.2 Random Forest Optimization

GridSearchCV was applied using a focused yet effective hyperparameter grid to optimize the Random Forest model. The parameters tested included `n_estimators` [50, 100], `max_depth` [20, 25], `min_samples_leaf` [1, 2], and `max_features` ['sqrt', 'log2']. The best configuration identified was:

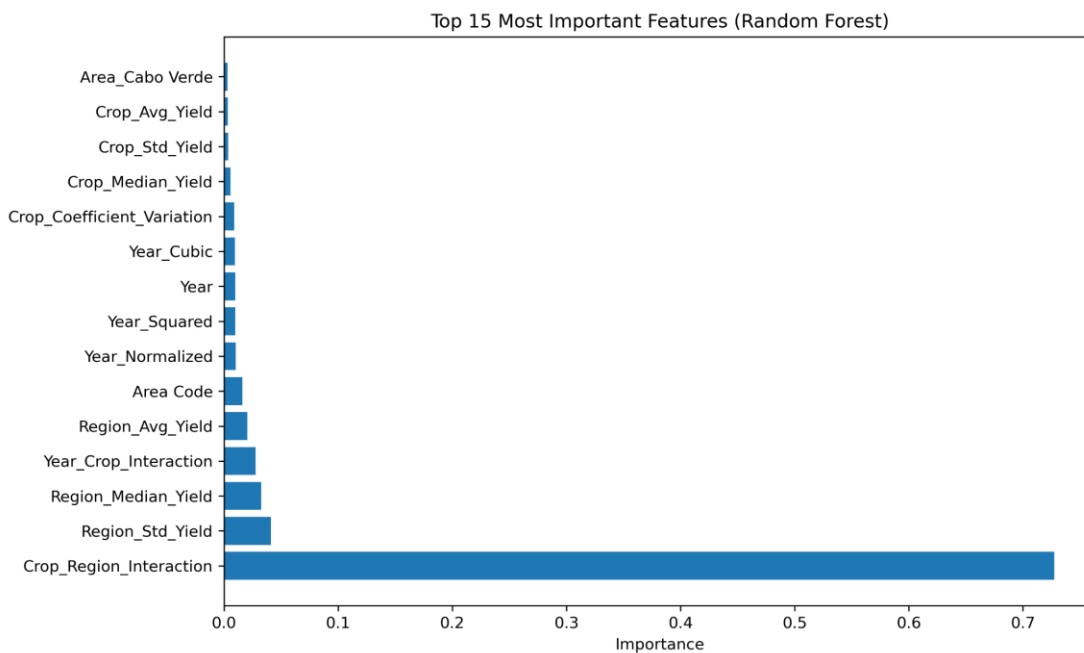
$n_estimators = 100$, $min_samples_leaf = 1$, $max_features = 'sqrt'$, and $max_depth = 25$. The optimized model achieved an RMSE of 0.17, MAE of 0.10, and an R^2 score of 0.97.

Model	RMSE (log)	MAE (log)	R^2 Score
Random Forest	0.17	0.10	0.9721

Optimized Random Forest reduces prediction error by over 50% compared to baseline

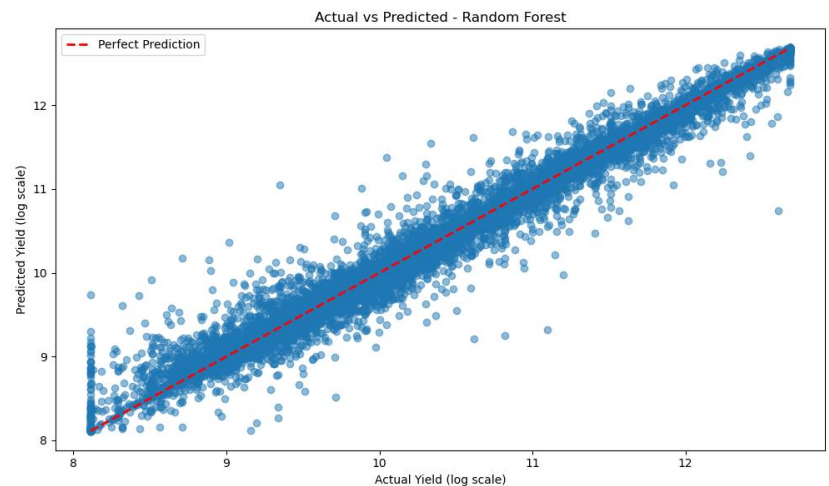
10. Feature Importance

Top 15 features were determined by optimized Random Forest.



11. Results Visualization and Interpretation

Strong linear correlation between actual and predicted values. Minimal systematic bias (points cluster around perfect prediction line). Some dispersion at extreme values (expected with log transformation).



12. Discussion and Limitations

12.1 Strengths of the Approach

1. **Comprehensive Feature Engineering:** Created 224 features from 4 base columns
2. **Multiple Model Comparison:** Evaluated diverse algorithms objectively
3. **Rigorous Optimization:** GridSearchCV with cross-validation
4. **Strong Performance:** $R^2 = 0.945$ exceeds typical agricultural prediction benchmarks
5. **Interpretability:** Feature importance provides actionable insights

12.2 Limitations and Constraints

The crop yield prediction model faces several limitations. From a **data perspective**, key variables such as weather (temperature, rainfall, humidity), soil characteristics (fertility, pH, composition), seasonal patterns, and management practices (irrigation, fertilization, pest control) were not included, which may reduce predictive accuracy. **Model limitations** include historical bias, as it assumes future trends mirror past patterns, regional specificity that may limit generalization to new areas, and variability in performance for rare crop types or long-term temporal extrapolation. Additionally, **computational constraints** arise due to the high-dimensional feature space (224 features), making GridSearchCV training resource-intensive and potentially challenging for real-time, large-scale deployment.

13. Conclusions

This project successfully developed a high-performance crop yield prediction system, achieving an **R^2 score of 0.945** and an **RMSE of 8,945 hg/ha**, demonstrating strong predictive capability. The pipeline—from raw data preprocessing and feature engineering to model training and evaluation—is fully operational, producing interpretable results through feature importance analysis and robust validation via cross-validation. In practical terms, the system can help **farmers** estimate expected yields, compare performance against regional averages, and identify underperforming crops for intervention. **Governments** can use it to forecast national production, plan import/export policies, and allocate subsidies efficiently, while **supply chain managers** can optimize inventory, anticipate market fluctuations, and negotiate contracts with data-driven insights. For production deployment, the **Optimized Random Forest** is recommended due to its superior accuracy, fast prediction time, robustness to outliers and missing data, and interpretability, all without requiring complex computational infrastructure.

14. References and Resources

14.1 Dataset

Food and Agriculture Organization (FAO). (2016). *Crop production statistics*.
<http://www.fao.org/faostat/>

14.2 Libraries and Tools

- Scikit-learn Documentation: <https://scikit-learn.org/>
- TensorFlow/Keras Documentation: <https://www.tensorflow.org/>
- Towards Data Science: Agricultural ML Articles